Thomas A. Berrueta

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Postdoctoral scholar working on ***embodied learning in safety-critical systems***. My research combines insights from reinforcement learning, optimal control, and statistical physics to develop principled methods that leverage powerful data-driven models to let agents reason about their environment interactions in real-time.

Experience

IEEE Robotics and Automation Letters, *Associate Editor* 2025 – Present

- Handling manuscripts on Aerial & Field Robotics and Theoretical Foundations.

California Institute of Technology, *Postdoctoral Scholar, Computing + Mathematical Sciences* 2024 – Present

- **Advisor:** Prof. Soon-Jo Chung.
- Technical lead of Caltech's Indy Autonomous Challenge autonomous IndyCar project.

Education

Northwestern University, *Ph.D., Mechanical Engineering*, Presidential Fellow 2017 – 2024

- **Advisor:** Prof. Todd D. Murphey.
- **Thesis:** *Robot Thermodynamics*

Northwestern University, *M.S., Mechanical Engineering* 2017 – 2019

- **Thesis:** *Data-Driven Symbolic Abstractions for Motion Analysis and Control*

Harvey Mudd College, *B.S., Engineering*, Graduated with Honors 2013 – 2017

Honors

- Northwestern University Presidential Fellowship (Highest Honor Attainable) 2022 – 2024
- Microsoft Future Leader in Robotics and AI 2024
- Schmidt Science Fellows Finalist 2023
- Microsoft Ada Lovelace Fellowship Finalist 2019
- Walter P. Murphy Fellowship 2017 – 2018
- Harvey S. Mudd Merit Scholarship 2013 – 2017
- Dean's List 2013 – 2017

Publications

Journal Papers (Indicates Equal Authorship):*

- **T. A. Berrueta**, A. Pinosky, T. D. Murphey, "Maximum diffusion reinforcement learning." *Nature Machine Intelligence*, vol. 6, no. 5, pp. 504-514 (2024).
- A. T. Taylor, **T. A. Berrueta**, A. Pinosky, T. D. Murphey, "Safe coverage for heterogeneous systems with limited connectivity." *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8866-8873 (2024).
- **T. A. Berrueta**, T. D. Murphey, R. L. Truby, "Materializing autonomy in soft robots across scales." *Advanced Intelligent Systems*, vol. 6, no. 2, 2300111 (2023).
- **T. A. Berrueta***, J. F. Yang*, A. M. Brooks, A. T. Liu, G. Zhang, D. G. Medrano, S. Yang, V. B. Koman, P. Chvykov, M. Z. Miskin, T. D. Murphey, M. S. Strano, "Emergent microrobotic oscillators via asymmetry-induced order." *Nature Communications*, vol. 13, 5734 (2022).
- J. F. Yang, A. T. Liu, **T. A. Berrueta**, G. Zhang, A. M. Brooks, V. B. Koman, S. Yang, X. Gong, T. D. Murphey, M. S. Strano, "Memristor circuits for colloidal robotics: Temporal access to memory, sensing, and actuation." *Advanced Intelligent Systems*, vol. 4, no. 4, 2100205 (2022).
- P. Chvykov, **T. A. Berrueta**, A. Vardhan, W. Savoie, A. Samland, T. D. Murphey, K. Wiesenfeld, D. I. Goldman, J. L. England, "Low rattling: A predictive principle for self-organization in active collectives." *Science*, vol. 371, no. 6524, pp. 90-95 (2021).

- A. T. Taylor, **T. A. Berrueta**, T. D. Murphey, "Active learning in robotics: A review of control principles." *Mechatronics*, vol. 77, 102576 (2021).
- W. Savoie, **T. A. Berrueta**, Z. Jackson, A. Pervan, R. Warkentin, S. Li, T. D. Murphey, K. Wiesenfeld, D. I. Goldman, "A robot made of robots: Emergent transport and control of a smarticle ensemble." *Science Robotics*, vol. 4, no. 34 (2019).
- **T. A. Berrueta**, A. Pervan, K. Fitzsimons, T. D. Murphey, "Dynamical system segmentation for information measures in motion." *IEEE Robotics and Automation Letters*, vol. 4, no. 1, pp. 169-176 (2019).

Conference Papers (Indicates Equal Authorship):*

- J. Ibrahim, **T. A. Berrueta**, S. J. Chung, Fred Y. Hadaegh,"Wasserstein contractive filtering for nonlinear systems under non-Gaussian uncertainties." *2026 IFAC World Congress* (In Preparation).
- J. Lathrop, G. Hua, **T. A. Berrueta**, S. J. Chung, "Learning to optimize autonomous racing strategies with compound kernel exploration." *IEEE Robotics and Automation Letters* (In Review).
- D. Chen*, A. Zimermann*, **T. A. Berrueta**, S. J. Chung, Fred Y. Hadaegh, "Environment-aware learning of smooth GNSS covariance dynamics for autonomous racing." *IEEE International Conference on Robotics and Automation (ICRA)* (In Review).
- A. Pinosky, **T. A. Berrueta**, O. Li, T .D. Murphey, "Physical state exploration for reinforcement learning from scratch." *IEEE International Conference on Automation Science and Engineering (CASE)* (2025).
- S. G. Kumar, J. Ibrahim, **T. A. Berrueta**, S. J. Chung, Fred Y. Hadaegh, "Real-time learning based planning for autonomous rendezvous in space with actuator loss of effectiveness." *AAS Guidance, Navigation & Control Conference (GNC)* (2025).
- **T. A. Berrueta***, A. Q. Nilles*, A. Pervan*, T .D. Murphey, S. M. LaValle, "Information requirements of collision-based micromanipulation." *Proceedings of the Workshop on the Algorithmic Foundations of Robotics (WAFR)* (2020).
- A. Kalinowska, **T. A. Berrueta**, T .D. Murphey, "Data-driven gait segmentation for walking assistance in a lower-limb assistive device." *IEEE International Conference on Robotics and Automation (ICRA)* (2019).

Workshop Papers and Abstracts:

- **T. A. Berrueta**, N. Ranganathan, S. J. Chung, "Active learning of tire parameters for high-speed autonomous racing." *Robotics: Science and Systems*, (2025).
- A. T. Taylor, **T. A. Berrueta**, T. D. Murphey, "Emergent mechanism design via robot swarms." *Robotics: Science and Systems*, (2022).
- **T. A. Berrueta**, J. F. Yang, A. M. Brooks, A. T. Liu, M. S. Strano, T. D. Murphey, "Emergent beating in colloidal matter: Stabilization via symmetry breaking." *Bulletin of the American Physical Society*, (2022).
- A. T. Taylor, **T. A. Berrueta**, T. D. Murphey, "Optimizing the locomotion of a robotic active matter system of smarticles." *Bulletin of the American Physical Society*, (2021).
- A. Q. Nilles, A. Pervan, **T. A. Berrueta**, T. D. Murphey, "Controlling active Brownian particles with 'active billiard' particles." *Bulletin of the American Physical Society*, (2020).
- **T. A. Berrueta**, A. Pervan, T. D. Murphey, "Towards robust motion planning for synthetic cells in a circulatory system." *Robotics: Science and Systems*, (2019).

Book Chapters:

- **T. A. Berrueta**, I. Abraham, T. D. Murphey, "Experimental applications of the Koopman operator in active learning for control." *The Koopman Operator in Systems and Control*, Springer (2020).

Invited Talks

- **T. A. Berrueta**, "Robot learning on the edge: Online learning in hardware." *ELLIIT Robot Learning Symposium*, Lund University, Sweden, November 20th, 2025.
- **T. A. Berrueta**, "Inside the Minds Building The Robots That Will Run the World (Panel)." *TECH WEEK by Andreessen-Horowitz (a16z)*, Los Angeles, October 16th, 2025.
- **T. A. Berrueta**, "Towards transparent & reliable embodied reinforcement learning agents." *Microsoft Future Leaders in Robotics and AI Seminar Series*. University of Maryland, April 26th, 2024, YouTube Video

- **T. A. Berrueta**, “Robot thermodynamics: Making complex systems task-capable." *Gordon Research Conference* (GRC): Complex Active and Adaptive Material Systems, Ventura Beach Marriott, February 1st, 2023.
- **T. A. Berrueta**, “Engineering robotic active matter through nonequilibrium self-organization." *Gordon Research Seminar* (GRS): Emergent Phenomena in Active and Living Materials, Ventura Beach Marriott, January 29th, 2023.
- **T. A. Berrueta**, “Imprecision engineering: Lowering cost while increasing capability." *Presidential Fellows Lecture Series*, Northwestern University, October 20th, 2022.
- **T. A. Berrueta**, “Designing for emergence: Making materials ‘robotic’ with self-organization." *Center for Robotics and Biosystems Seminar Series*, Northwestern University, June 17th, 2022, YouTube Video.
- **T. A. Berrueta**, “Robot thermodynamics: Analysis, control, and design of complex systems." *Allen Discovery Center Invited Seminar*, Department of Biology, Tufts University, April 26th, 2022.
- **T. A. Berrueta**, “Online learning in physical systems." *SIAM Dynamical Systems*, Symposium on Leveraging Machine Learning for Dynamics and Control, May 26th, 2021, YouTube Video.
- **T. A. Berrueta**, “Low rattling: Predicting driven self-organization." *Center for Robotics and Biosystems Seminar Series*, Northwestern University, December 4th, 2020, YouTube Video.

Teaching

ME455: Active Learning in Robotics Co-Lecturer (2022, 2023, 2024)

- Lectured and designed teaching materials for graduate-level course (∼30 students) focusing on the closed-loop relationship between the between information needs and physical actions of a robot.
- Lectured for 30% of the semester’s classes in 2024 and provided guest lectures in 2022 and 2023.
- Developed brand new lecture materials spanning multiple units on topics such as probability theory, information theory, Bayesian inference, estimation, particle filters, optimal control, reinforcement learning, and deep learning architectures like CVAEs.

ME314: Theory of Machines – Dynamics Teaching Assistant (2019), Grader (2018, 2021)

- Lectured large senior-level course (∼70 students) on rigid body dynamics and Lagrangian mechanics.
- Redesigned class content and homeworks as it pivoted to a more programming-heavy curriculum.

Service

- **Reviewer:** *Nature Communications, Nature Scientific Reports, IEEE Transactions on Robotics (T-RO), IEEE Robotics and Automation Letters (RA-L), IEEE Transactions on Control Systems Technology (TCST), IEEE International Conference on Robotics and Automation (ICRA), IEEE International Conference on Automation Science and Engineering (CASE), IEEE International Conference on Intelligent Robots and Systems (IROS), IEEE American Control Conference (ACC).*
- **Session Chair:** *IEEE International Conference on Automation Science and Engineering (CASE)*

Leadership & Outreach

California Institute of Technology

- Technical lead and project manager of Caltech Racer (2025 – Present).
- Mentor to 4 Ph.D. students students as part of the Indy project (2024 – Present).
- Mentor and manager of 2 undergraduate summer research internship projects (2025).

Northwestern University

- Board member of the Northwestern Mechanical Engineering Graduate Student Society (2017 – 2019).
- Organizer of recruitment activities for incoming mechanical engineering graduate students (2018).
- Mentor to 8 Ph.D. and Masters students through the department mentorship program (2017 – 2024).
- Volunteer at the Chicago Museum of Science and Industry, teaching members of the public about robots and technology (2017 – 2024).

Harvey Mudd College

- Member of the Harvey Mudd College Entrepreneurial Network (2017).
- Mentor in the Harvey Mudd College mentorship program (2016 – 2017).

Industry Experience

Amazon Lab126 Device Test Engineering Intern (2017)

- Designed a robotic hardware platform for testing and evaluating the voice-responsiveness of the Amazon Echo.

Northrup Grumman Survivability Control Systems Intern (2016)

- Developed an infrared signature estimator model and worked on algorithms for minimizing aircraft IR exposure.

SpaceX Vehicle Engineering Intern (2014, 2015)

- Modelled and characterized dynamical flight environments experienced by the Falcon 9's M1D and MVacD rocket engines in order to anticipate failure modes and ensure mission success.

Press

- *Caltech Magazine*, "Caltech's Autonomous IndyCar Shows Promise at First US Road Course Challenge", 2025.
- *Caltech Magazine*, "Computer, Start Your Engine", 2025.
- *ArsTechnica*, "Exploration-focused training lets robotics AI immediately handle new tasks", 2024.
- *Northwestern News*, "This algorithm makes robots perform better", 2024.
- *Northwestern News*, "PhD Student Chosen as Presenter for Future Leaders in Robotics and AI Series", 2024.
- *Northwestern News*, "Chemistry flexes robotic arm without electronics," 2022.
- *MIT News*, "Tiny particles work together to do big things," 2022.
- *Northwestern News*, "Two Graduate Students Receive Presidential Fellowships," 2022.
- *Science (Podcast)*, "The uncertain future of North America's ash trees, and organizing robot swarms," 2021.
- *Gizmodo*, "Meet the Pint-Sized Robots that Spontaneously Dance," 2020.
- *Popular Mechanics*, "These Robots Literally Just Flap Their Wings. That's It. But the Army Loves Them.," 2019.
- *Scientific American*, "Prehistoric Suckers, Slapping Robots and Three Billion Birds Gone," 2019.
- *Science (News)*, "Watch a robot made of robots move around," 2019.